TASK 5: EXPLORATORY DATA ANALYSIS (EDA)

Titanic Dataset – Solved Practical with Python Code

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Course: B.Sc. Data Science

Task: 5 – Exploratory Data Analysis

Tools: Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

Dataset: Titanic Dataset

1. Objective

The objective is to explore the Titanic dataset using statistical summaries and visualizations, identify patterns, relationships, trends, missing values and anomalies, and summarize meaningful insights.

2. Dataset Description

The Titanic dataset contains passenger information including survival status, passenger class, sex, age, family-related variables, ticket, fare, cabin and port of embarkation. The main outcome variable is Survived.

3. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
%matplotlib inline
```

4. Load Dataset

```
df = pd.read_csv("Titanic-Dataset.csv")
display(df.head())
```

Observation: The first five passenger records are displayed.

5. Dataset Structure

```
print("Shape:", df.shape)
df.info()
print(df.columns.tolist())
```

Observation: info() shows the number of records, columns, data types and non-null values.

6. Descriptive Statistics

```
df.describe(include="all")
```

Observation: describe() provides central tendency, spread, quartiles and other summary statistics.

7. Missing Values

```
print(df.isnull().sum())
missing_pct = (df.isnull().sum()/len(df)*100).sort_values(ascending=False)
print(missing_pct)
```

Observation: Cabin commonly contains many missing values in the Titanic dataset, while Age also has missing observations. Missing-value treatment should be based on the analysis objective.

8. Duplicates and Value Counts

```
print("Duplicate rows:", df.duplicated().sum())
print(df["Survived"].value_counts())
print(df["Sex"].value_counts())
print(df["Pclass"].value_counts())
```

Observation: Value counts show the distribution of survival, gender and passenger class. More passengers did not survive than survived.

9. Data Cleaning

```
eda = df.copy()
eda["Age"] = eda["Age"].fillna(eda["Age"].median())
if eda["Embarked"].isnull().any():
    eda["Embarked"] = eda["Embarked"].fillna(eda["Embarked"].mode()[0])
if "Cabin" in eda.columns:
    eda = eda.drop(columns=["Cabin"])
print(eda.isnull().sum())
```

Observation: Median imputation is used for Age, mode imputation for Embarked, and Cabin is removed for this basic EDA because of extensive missingness.

10. Univariate Analysis

10.1 Survival Distribution

```
sns.countplot(data=eda, x="Survived")
plt.title("Survival Distribution")
plt.xlabel("Survived (0=No, 1=Yes)")
plt.show()
```

Observation: Non-survivors are more numerous than survivors.

10.2 Age Histogram

```
sns.histplot(data=eda, x="Age", bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```

Observation: Passenger ages are concentrated mainly among younger and middle-aged passengers, with fewer very old passengers.

10.3 Fare Boxplot

```
sns.boxplot(data=eda, x="Fare")
plt.title("Fare Distribution")
plt.show()
```

Observation: Fare is right-skewed and contains high-value outliers.

11. Bivariate Analysis

11.1 Survival by Gender

```
sns.countplot(data=eda, x="Sex", hue="Survived")
plt.title("Survival by Gender")
plt.show()
```

Observation: Female passengers have a substantially higher survival rate than male passengers.

11.2 Survival by Passenger Class

```
sns.countplot(data=eda, x="Pclass", hue="Survived")
plt.title("Survival by Passenger Class")
plt.show()
```

Observation: First-class passengers generally show better survival outcomes than second- and third-class passengers.

11.3 Age vs Fare

```
sns.scatterplot(data=eda, x="Age", y="Fare", hue="Survived", alpha=.7)
plt.title("Age vs Fare by Survival")
plt.show()
```

Observation: Most fares are relatively low, with a smaller number of high-fare observations. Survival varies across passenger characteristics.

12. Pairplot

```
pair_cols = ["Survived","Pclass","Age","SibSp","Parch","Fare"]
sns.pairplot(eda[pair_cols], hue="Survived", corner=True)
plt.show()
```

Observation: The pairplot displays multiple pairwise relationships and helps identify clusters, trends and unusual observations.

13. Correlation Heatmap

```
corr = eda[["Survived","Pclass","Age","SibSp","Parch","Fare"]].corr()
plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```

Observation: The heatmap summarizes linear correlations. Survived is associated positively with Fare and negatively with Pclass. Correlation does not imply causation.

14. Survival Rate

```
print(eda.groupby("Sex")["Survived"].mean())
print(eda.groupby("Pclass")["Survived"].mean())
```

Observation: Grouped means give survival rates and clearly show differences by gender and passenger class.

15. Outlier Detection using IQR

```
Q1 = eda["Fare"].quantile(.25)
Q3 = eda["Fare"].quantile(.75)
IQR = Q3 - Q1
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
outliers = eda[(eda["Fare"] < lower) | (eda["Fare"] > upper)]
print("Fare outliers:", len(outliers))
```

Observation: The IQR method identifies unusually high or low fare observations. Outliers should be investigated before removing them.

16. Key Findings

- More passengers did not survive than survived.
- Female passengers had a much higher survival rate than male passengers.
- Passenger class is strongly associated with survival.
- First-class passengers generally had better survival outcomes.
- Fare is right-skewed and contains high-value outliers.

- Age contains missing values and can be median-imputed for basic EDA.
- Pairplots reveal pairwise patterns, while heatmaps summarize correlations.
- EDA identifies data-quality issues and relationships before statistical modelling or machine learning.

17. Conclusion

EDA was successfully performed on the Titanic dataset using Pandas, Matplotlib and Seaborn. Descriptive statistics, missing-value analysis, histograms, boxplots, countplots, scatterplots, pairplots and a correlation heatmap were used. Gender, passenger class and fare show important associations with survival. The analysis provides a foundation for further statistical analysis and predictive modelling.

18. Interview Questions and Answers

1. What is EDA and why is it important?

EDA is the process of exploring a dataset using statistics and visualizations. It helps find missing values, outliers, distributions, relationships, trends and data-quality problems before modelling.

2. Which plots do you use to check correlation?

Scatterplots are useful for two numerical variables. A correlation heatmap shows correlations among many numerical variables. Pairplots show several pairwise relationships.

3. How do you handle skewed data?

Identify skewness using histograms, boxplots or a skewness statistic. Depending on the purpose, use log, square-root or another transformation, robust statistics, or a model suitable for skewed data.

4. How do you detect multicollinearity?

Use a correlation matrix and Variance Inflation Factor (VIF). High predictor correlation or high VIF indicates possible multicollinearity.

5. What are univariate, bivariate and multivariate analyses?

Univariate studies one variable, bivariate studies two variables, and multivariate studies three or more variables together.

6. Difference between heatmap and pairplot?

A heatmap displays values in a matrix using color, commonly correlations. A pairplot displays scatterplots and distributions for several variables.

7. How do you summarize your insights?

State the important patterns, relationships, trends, outliers and data-quality issues in clear bullet points, supported by statistics or visualizations.

19. Complete Jupyter Notebook Code

```
# TASK 5 – EDA TITANIC DATASET

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set_theme(style="whitegrid")

df = pd.read_csv("Titanic-Dataset.csv")
display(df.head())
print(df.shape)
df.info()
display(df.describe(include="all"))

print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

for col in ["Survived","Sex","Pclass","Embarked"]:
    if col in df.columns:
        print("\n", col)
        print(df[col].value_counts(dropna=False))

eda = df.copy()
eda["Age"] = eda["Age"].fillna(eda["Age"].median())
if eda["Embarked"].isnull().any():
    eda["Embarked"] = eda["Embarked"].fillna(eda["Embarked"].mode()[0])
if "Cabin" in eda.columns:
    eda = eda.drop(columns=["Cabin"])

# Univariate
sns.countplot(data=eda, x="Survived"); plt.title("Survival Distribution"); plt.show()
sns.histplot(data=eda, x="Age", bins=30, kde=True); plt.title("Age Distribution"); plt.show()
sns.boxplot(data=eda, x="Fare"); plt.title("Fare Distribution"); plt.show()

# Bivariate
sns.countplot(data=eda, x="Sex", hue="Survived"); plt.title("Survival by Gender"); plt.show()
sns.countplot(data=eda, x="Pclass", hue="Survived"); plt.title("Survival by Class"); plt.show()
sns.scatterplot(data=eda, x="Age", y="Fare", hue="Survived"); plt.title("Age vs Fare"); plt.show()

# Multivariate
pair_cols=["Survived","Pclass","Age","SibSp","Parch","Fare"]
sns.pairplot(eda[pair_cols], hue="Survived", corner=True); plt.show()

corr=eda[pair_cols].corr()
plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()

# Survival rates
print(eda.groupby("Sex")["Survived"].mean())
print(eda.groupby("Pclass")["Survived"].mean())

# IQR outliers
Q1=eda["Fare"].quantile(.25); Q3=eda["Fare"].quantile(.75)
IQR=Q3-Q1
outliers=eda[(eda["Fare"]<Q1-1.5*IQR)|(eda["Fare"]>Q3+1.5*IQR)]
print("Fare outliers:",len(outliers))
```